

Bernard K. Appiah

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PROFESSIONAL SUMMARY

Emerging biostatistician with two years of experience in statistical modeling, public-health data analysis, and quantitative research. Skilled in survival methods, machine learning, and regression, with master's-level training in advanced modeling, statistical inference, and high-dimensional data analysis. Brings hands-on experience in teaching, statistical consulting, and field data collection, with a proven ability to communicate technical results to non-statistical audiences and to collaborate effectively across disciplines and cultures. Driven to pursue advanced research in statistical methodology and to develop methods that address real-world scientific and public health problems.

EDUCATION

- University of Minnesota Duluth (UMD)** **Duluth, MN**
MS in Statistics; GPA: 4.0/4.0 **August 2025 - Present**
 - Research: Variable Screening Techniques for Survival Analysis
 - Advisor: Prof. Dananjani. M. Liyanage
- University of Ghana** **Accra, Ghana**
B.sc Mathematics and Statistics (Double Major); GPA: 3.64/4.00 **September 2019 - September 2023**
 - Research: Application of Machine Learning Models to Identify Malaria Risk Factors Among Pregnant Women in Ghana
 - Advisor: Dr. Gabriel Kallah-Dagadu
 - First Class Honors

TEACHING EXPERIENCE

- Department of Mathematics & Statistics, University of Minnesota Duluth** **Duluth, MN**
Teaching Assistant **August 2025– Present**
 - Designed and delivered 225+ lectures for Calculus I (MATH 1296), Calculus II (MATH 1297), Calculus III (MATH 1298), and Differential Equations (MATH 3280), reaching 300+ undergraduates across 3 semesters – Fall 2025, Spring 2026, and Fall 2026.
 - Offered one-on-one support during office hours, working closely with 20–35 students per section to tackle challenging problems and build confidence in the subject matter.
 - Provided detailed, constructive feedback on student assignments and exams, helping students identify areas for improvement while celebrating their progress.
 - Conducted targeted review sessions and lab sessions using Mathematica and Python, reinforcing key concepts and strengthening students' computational skills.
- Department of Actuarial Science and Statistics, University of Ghana** **Accra, Ghana**
Teaching Assistant & Research Assistant **November 2023 – December August 2024**
 - Led instruction in STAT 333, STAT 334, and STAT 445 for 400+ undergraduates who found inference and regression the hardest content in the major, covering both lecture and discussion sections.

- Provided 10+ hours weekly of one-on-one R and SPSS guidance for students unable to translate coursework theory into working code, serving as the department's main applied-computing support channel.
- Supported faculty research spanning multiple concurrent projects by gathering and cleaning raw datasets that arrived incomplete and inconsistently formatted.
- Ran analyses in R and SPSS on those cleaned datasets, producing the summaries and model output that fed directly into faculty research objectives.

• **Ghana Baseline Evaluation Survey, University of Ghana**
Enumerator and Data Analyst

Accra, Ghana
January 2024 - February 2024

- Conducted door-to-door surveys to collect data on the livelihood of respondents in various communities in the Eastern region.
- Analyzed the survey results using statistical software to help make informed decisions.
- Maintained detailed records of survey progress, ensuring accurate and complete documentation.

RESEARCH EXPERIENCE

Manuscripts Under Review

- **Appiah, B. K.**, Incoom-Koomson, K., Bonney, S., Boye, M., & Kallah-Dagadu, G. (2026). Application of Machine Learning Models to Identify the Risk Factors of the Incidence of Malaria Among Pregnant Women in Ghana ([Link](#)).

Graduate Thesis (in progress)

Variable Screening Techniques for Survival Analysis

- Developing p-value based screening procedures for identifying influential covariates in high-dimensional censored time-to-event data.
- Evaluating performance through simulation studies across varying dimensionality, censoring rates, and correlation structures, benchmarked against penalized Cox regression.
- Applying the methods to a real clinical survival dataset.

Undergraduate Research

Application of Machine Learning Models to Identify the Risk Factors of the Incidence of Malaria Among Pregnant Women in Ghana ([Link](#))

- Applied Random Forest, Decision Trees, KNN, Naive Bayes, and Neural Networks to 2019 Ghana Malaria Indicator Survey data to identify risk factors for malaria incidence among pregnant women.

Academic Projects

- *Comparative Statistical and Machine Learning Approaches for Life Expectancy Prediction in Africa*. Modeled determinants across 54 countries (832 country-year observations); benchmarked via stratified train-test splits, repeated 10-fold cross-validation, and bootstrapping, with Random Forest and Gradient Boosting outperforming traditional regression ([Link](#)).
- *Modeling Life Expectancy in Africa Using Socioeconomic and Health Indicators*. Multiple linear regression identifying adult mortality and HIV incidence as significant drivers ([Link](#)).

- *A Project Proposal to Investigate the Social, Economic and Legal Aspects of the Galamsey Problem in Ghana.* Proposed a research design covering environmental degradation, socioeconomic impact on mining communities, and the regulatory framework.

Presentations

- *Application of Machine Learning Models to Identify Malaria Risk Factors Among Pregnant Women in Ghana.* Poster Presentation, Swenson College of Science and Engineering (SCSE) Student Research Day, University of Minnesota Duluth. February 2026.
- *Modeling Life Expectancy in Africa Using Socioeconomic and Health Indicators.* Class Presentation, University of Minnesota Duluth. December 2025.

Talks

- *Application of Machine Learning Models to Identify Malaria Risk Factors Among Pregnant Women in Ghana.* Departmental Seminar, University of Ghana, Accra, Ghana. November 2023.
- *Understanding the Least Squares Method using Linear Algebra.* Class Presentation, University of Minnesota Duluth. April 2026.
- *Cramér–Rao Lower Bound and Efficiency of Estimators.* Class Presentation, University of Minnesota Duluth. September 2025.
- *A Statistical Look at Barcelona's Performance: Last Season vs This Season.* Class Presentation, University of Minnesota Duluth. August 2025.

AWARDS AND HONORS

- 2026 – Student Research Grant, University of Minnesota Duluth
- 2026 – Fostering Diversity in Biostatistics Workshop Funding, ENAR
- 2023 – Winner, Mathematics Society Quiz Competition, University of Ghana
- 2023 – Winner, Statistics and Actuarial Science Quiz Competition

ACADEMIC SERVICE, LEADERSHIP, & VOLUNTEERISM

- 2021 – Member, Academic Committee, University of Ghana Actuarial and Statistics Society
- 2020 – Member, The Mathematics Society, Department of Mathematics, University of Ghana

NON-DEGREE EDUCATION & CERTIFICATIONS

- Exploratory Data Analysis in R – DataCamp
- Introduction to Regression in R – DataCamp
- Survival Analysis in R – DataCamp

HIGHLIGHTED SKILLS

- **Soft Skills:** Collaboration, Leadership, Analytical Thinking, and Effective Communication
- **Technical Skills:** Statistical Modeling (R, Python, SPSS), $\LaTeX$, GitHub and Microsoft 365

PROFESSIONAL MEMBER ASSOCIATION

- American Statistical Association
- American Mathematical Society
- Institute of Mathematical Statistics
- Eastern North American Region of the International Biometric Society